SEEING CHANGE FROM ORBIT

Mangrove Segmentation and Change Analysis

Technical report for the Eastern Mangroves using Sentinel-2 imagery, QGIS, and a six-channel U-Net.



Preliminary 2025 U-Net mangrove mask displayed over the Sentinel-2 image. The green overlay is a model output that still requires manual review.

2018-2025	6 channels	F1 0.814
Study period	RGB, NIR, NDVI, NDWI	Preliminary geographic test

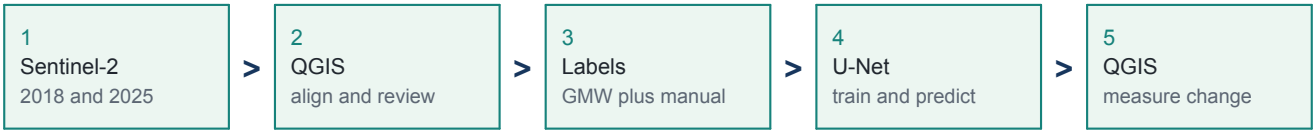
Status: preliminary technical workflow. Ecological conclusions remain preliminary until the held-out reference polygons are independently reviewed.

1 Project scope and workflow

The project applies a U-Net-based mangrove segmentation workflow to Sentinel-2 imagery. QGIS provides geospatial preparation, reference review, and map analysis; Python and PyTorch provide patch generation, model training, prediction, and evaluation.

Workflow component	Implementation
Mangrove segmentation model	A compact six-channel U-Net, training script, checkpoint, and prediction pipeline.
Temporal analysis	2018 and 2025 predictions, a change raster, geographic validation, and test metrics.
Ground-reference data	Open Global Mangrove Watch 2018 and 2025 references plus an editable QGIS review layer.
Project outputs	A validated 14-layer QGIS project, figures, documentation, and reproducible commands.

The workflow



How QGIS and U-Net relate

The tools are complementary. QGIS manages coordinates, imagery, labels, visual inspection, and area measurement. The U-Net learns the spectral pattern of mangroves from image patches. Its output returns to QGIS as georeferenced rasters for review and mapping.

QGIS	Python and PyTorch
Open, align, style, and compare satellite layers	Create training patches and six-channel tensors
Inspect and correct reference polygons	Train the U-Net and calculate loss
Display predicted gain, loss, and stable mangroves	Generate probability masks and performance metrics
Measure area and produce final maps	Repeat experiments reproducibly

2 Study data and image preparation

The analysis uses two Sentinel-2 Level-2A scenes from the same MGRS tile. The acquisition dates are only four calendar days apart seasonally, which reduces seasonal mismatch. Both images were cropped and resampled onto the same 10 metre grid in WGS 84 UTM zone 40N.

Role	Scene	Date	Cloud
Before	S2A_40RBN_20181105_1_L2A	5 Nov 2018	0.36%
After	S2C_40RBN_20251101_0_L2A	1 Nov 2025	0.10%

Six model inputs

Input	Purpose
Blue	Water, haze, and visible-surface information
Green	Vegetation and water contrast; used in NDWI
Red	Visible vegetation absorption; used in NDVI
Near-infrared	Strong vegetation response and shoreline contrast
NDVI	(NIR - red) / (NIR + red), highlighting vegetation greenness
NDWI	(green - NIR) / (green + NIR), highlighting water and wet surfaces

Reference labels

Global Mangrove Watch version 4.1.12 supplied 90 clipped polygons for 2018 and 83 for 2025. These polygons are useful starting references, but they are deliberately labeled unreviewed in QGIS. Final test polygons must be manually checked against both images.

- All rasters align to an 828 by 902 pixel analysis grid.
- The visible and near-infrared inputs use 10 metre pixels.
- Cloud, shadow, and invalid pixels are screened using Sentinel-2 scene classification.
- Geographic blocks, not random neighboring patches, define training, validation, and test splits.

3 Understanding the 2018 to 2025 result figure

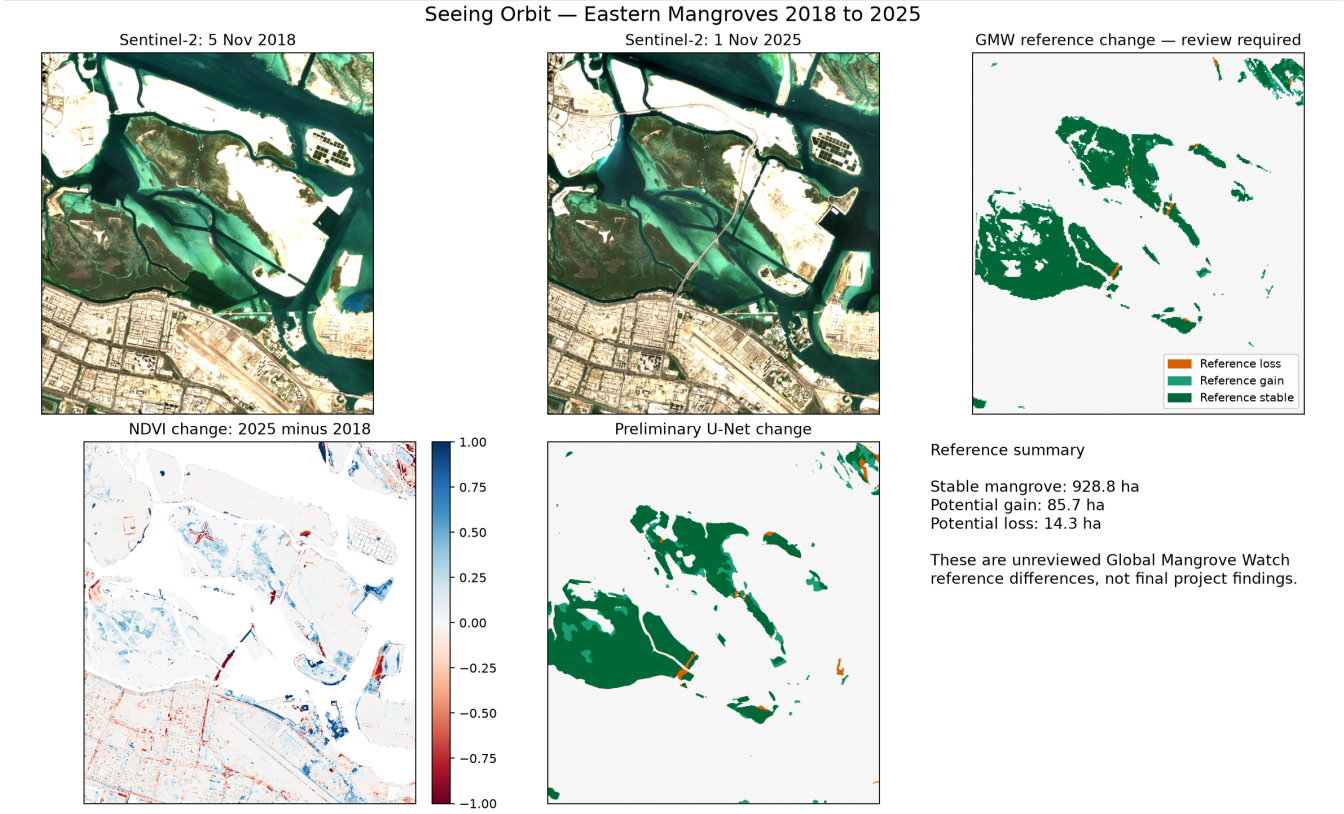


Figure 1. Sentinel-2 imagery, reference change, NDVI change, preliminary U-Net change, and unreviewed reference area summary for the Eastern Mangroves.

Panel explanation

- Top left and top centre: the same study area in 2018 and 2025. These true-colour views support visual confirmation.
- Top right: Global Mangrove Watch reference change. Dark green is stable reference mangrove, teal is potential gain, and orange is potential loss.
- Bottom left: NDVI change. Blue means a higher vegetation signal in 2025; red means a lower vegetation signal.
- Bottom centre: preliminary U-Net change. It applies the same gain, loss, and stable categories to the model masks.
- Bottom right: reference areas. These values are screening results, not final ecological findings.

The large visible construction changes are important sources of confusion. Bright reclaimed land, shallow water, tide, sediment, and narrow shoreline vegetation can all produce spectral differences that resemble mangrove change. Every proposed gain or loss therefore requires QGIS review.

4 Using the prepared QGIS project



Figure 2. The prepared QGIS project with the preliminary 2025 model mask in green. The orange boundary shows the study area.

Review procedure

- Open qgis/seeing-orbit-mangroves-2018-2025.qgs.
- Toggle True colour 2018-11-05 and True colour 2025-11-01 to compare the same location.
- Turn on the GMW 2018 and 2025 reference layers and inspect their boundaries.
- Turn on U-Net mangrove change 2018 to 2025 to see stable, gain, and loss predictions.
- Edit Manual mangrove review edit this layer and record class, confidence, reviewer, evidence, and notes.
- Save the GeoPackage frequently and keep the held-out test blocks independent from model training.

Key layers

Layer	Use
True colour 2018 and 2025	Primary visual comparison
GMW reference unreviewed	Starting reference boundaries, not final truth
Manual mangrove review	Editable reviewer decisions
NDVI and NDWI differences	Supporting vegetation and water evidence
U-Net predictions	Model extent for each date
U-Net mangrove change	Stable, gain, and loss comparison

5 U-Net model and evaluation

The model segments each date independently. It receives six aligned channels and produces one mangrove-probability map. The 2018 and 2025 binary masks are then compared pixel by pixel to classify stable mangrove, potential gain, and potential loss.

Model sequence

1 Sentinel-2 2018 and 2025 > 2 QGIS align and review > 3 Labels GMW plus manual > 4 U-Net train and predict > 5 QGIS measure change

Training configuration

Setting	Value
Architecture	Compact U-Net with 24 base channels
Training samples	38 geographic patches
Validation samples	8 geographic patches
Test samples	10 geographic patches
Patch size	256 by 256 pixels
Loss	Class-weighted binary cross-entropy plus soft Dice
Training	10 epochs, seed 42, Apple Metal acceleration
Saved checkpoint	Epoch 9 selected by validation F1

Preliminary reference-label agreement

Split	Precision	Recall	F1	IoU
Validation	0.746	0.965	0.841	0.726
Test	0.744	0.898	0.814	0.686

These scores measure agreement with unreviewed Global Mangrove Watch masks. They demonstrate that the technical pipeline works, but they are not final accuracy measurements against independent, manually adjudicated ground truth.

6 Interpreting the preliminary findings

Source	Stable	Gain	Loss	Net
GMW reference	928.8 ha	85.7 ha	14.3 ha	+71.4 ha
Preliminary U-Net	919.3 ha	149.5 ha	31.6 ha	+117.9 ha

What the disagreement means

The model marks more potential gain and loss than the reference dataset. This does not prove that the model found additional real change. The difference may come from false positives, boundary detail, actual local change omitted from the reference, or inconsistent treatment of water, bare soil, construction, and narrow mangrove edges. The disagreement defines the areas that should be reviewed first in QGIS.

Main limitations

- Sentinel-2 pixels are 10 metres, so narrow or fragmented mangrove strips may be mixed with water or soil.
- Different tidal state, water turbidity, haze, and shadows can change NDVI and NDWI without ecological change.
- Global Mangrove Watch is an external mapped product, not project-specific manually verified truth.
- The study area is small; 56 patches are sufficient for a pipeline demonstration but not a broad regional claim.
- The same model architecture and threshold must be evaluated across additional Abu Dhabi mangrove areas.

Required scientific gate

- Review every test block independently and record confidence.
- Have a second team member review all final test polygons.
- Adjudicate disagreements without using model predictions as the deciding evidence.
- Re-export the corrected masks and calculate final precision, recall, F1, IoU, gain, and loss.
- Report confidence-3 results separately from confidence-2-plus-3 results.

7 Reproducibility, files, and references

Artifact	Location
QGIS project	qgis/seeing-orbit-mangroves-2018-2025.qgs
Configuration	configs/eastern_mangroves_2018_2025.yaml
Reference labels	data/labels/mangrove_2018_2025.gpkg
Model checkpoint	checkpoints/mangrove_unet_2018_2025.pt
Predictions	data/derived/2018_2025/model/
Detailed workflow	README.md and docs/experiment-protocol.md
Results summary	reports/preliminary-unet-results.md

Reproduce the workflow

```
make setup-ml
make mangrove-data
make mangrove-labels
make mangrove-patches
.venv/bin/python scripts/train_mangrove_unet.py --epochs 20
make mangrove-predict
make mangrove-evaluate
```

References

1. Element 84. Earth Search STAC API. Public catalogue used to locate the Sentinel-2 scenes.

2. Bunting, P. et al. Global Mangrove Watch Timeseries of Mangrove Extent version 4.1.12. Zenodo.

3. QGIS Development Team. QGIS Geographic Information System.

4. PyTorch Foundation. PyTorch.

Free-use statement

This workflow used QGIS, Python, PyTorch, public Sentinel-2 imagery, and the open Global Mangrove Watch dataset. No paid API, cloud-compute subscription, trial, payment information, or paid model access was used.

Final validation requires manual review of the held-out polygons before the area-change values or model scores are presented as final results.